

Q14 In order to get destructive interference at Q, the path difference of waves moving between P and Q should be $(m - 0.5) \lambda$.

$$\text{Hence, path difference} = PS_1Q - PS_2Q = (l_1 + l_3) - (l_2 + l_4) = \left(\frac{2m - 1}{2} \right) \lambda$$

Answer: D

Q15 Using Rayleigh Criterion